

ML-Wheat Multi-Scale Feature Documentation

Variables in `wheat_multiscale_features_{year}.parquet`

Kelly Trinh

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1 Overview

The file `wheat_multiscale_features_{year}.parquet` contains one row per wheat-growing grid cell ($\sim 26,736$ cells per year) with features extracted at three temporal resolutions:

1. **Weekly** — $60 \text{ slots} \times 16 \text{ variables} = 960 \text{ features}$
2. **Monthly** — $15 \text{ slots} \times 16 \text{ variables} = 240 \text{ features}$
3. **Crop-seasonal** — $6 \text{ windows} \times 16 \text{ variables} + \sim 25 \text{ extras} \approx 121 \text{ features}$
4. **Static** — 9 identifiers and soil properties

Total: $\sim 1,320$ s features per grid cell per year.

All features are derived from four SILO climate inputs (tmin, tmax, rain, radn), gridded soil moisture (s0), and static APSIM soil properties. The APSIM-Wheat documentation (v7.5 R3008) is referenced throughout with equation numbers.

1.1 Data Sources

Source	Variables	Resolution
SILO climate NetCDF	tmin, tmax, rain, radn	Daily, 0.05°
Soil moisture (s0)	Plant-available water (mm)	Daily, gridded
APSIM soil XML	PAWC, pH, mineral N, depth	Static per soil type
Gridded soil map	Dominant soil ID per cell	Static per cell
DEWS yield NetCDF	Wheat yield (kg/ha $\rightarrow$ t/ha)	Annual

1.2 Sowing Date Determination

Sowing dates are computed per cell per year using DEWS rules (not fixed):

- **Northern** (lat > -32.24 , lon > 140.99): rain $\geq 15 \text{ mm}/3 \text{ days}$ AND PAW $\geq 30 \text{ mm}$, Apr 26–Jul 31
- **Southern** (lat < -32.25 , lon > 129): rain $\geq 15 \text{ mm}/3 \text{ days}$, Apr 26–Jul 31
- **Western** (lon ≤ 129): rain $\geq 15 \text{ mm}/3 \text{ days}$, Apr 1–Jul 31
- Wet-soil delay: PAW $< 95\%$ of PAWC. Default sow Jul 31 if no criteria met.

2 Static Features (9 columns)

These are time-invariant identifiers and soil properties.

Column	Unit	Description
lat	$^\circ\text{S}$	Latitude of grid cell
lon	$^\circ\text{E}$	Longitude of grid cell
sowing_doy	day	Day-of-year of sowing (NaN if not sown)
no_sow	0/1	1 if no sowing occurred
pawc_0_30_mm	mm	Plant Available Water Capacity, 0–30 cm depth (APSIM Eq 70, 89–90)
pawc_0_60_mm	mm	PAWC, 0–60 cm depth
ph_0_30	—	Depth-weighted mean soil pH, 0–30 cm

Column	Unit	Description
minN_0_30	mg/kg	Mineral nitrogen ($\text{NO}_3 + \text{NH}_4$), 0–30 cm (relates to Eq 98–99)
profile_depth_cm	cm	Total soil profile depth (constrains root growth, Section 9.1)
wheat_yield	t/ha	APSIM-simulated yield (target variable)
year	—	Calendar year

3 Temporal Slot Variables (16 per slot)

The same 16 variables are computed for every temporal slot (weekly, monthly, and crop-seasonal). Each variable name is formed as {variable}_{slot_name}, e.g. `rain_sum_w3` or `fasw_mean_mpre2`.

Variable	Unit	Description and APSIM Reference
rain_sum	mm	Total rainfall. Water input for the soil water balance (Section 11).
rain_days	days	Number of days with rainfall ≥ 1 mm. Rainfall frequency pattern.
rad_mean	MJ/m ² /d	Mean daily solar radiation. Drives biomass accumulation via RUE (Eq 13–16).
rad_sum	MJ/m ²	Cumulative radiation. Total energy available for photosynthesis.
tmean	°C	Mean daily temperature $(T_{\text{max}} + T_{\text{min}})/2$. General thermal environment.
tmax_max	°C	Maximum of daily T_{max} . Detects heat extremes ($>34^\circ\text{C}$ triggers senescence, Eq 81).
tmin_min	°C	Minimum of daily T_{min} . Detects frost events ($<0^\circ\text{C}$, Eq 80).
diurnal	°C	Mean diurnal temperature range ($T_{\text{max}} - T_{\text{min}}$). Influences VPD and vernalisation (Eq 9).
tt_sum	°Cd	Thermal time accumulated (Eq 4). Cardinal temperatures: 0/26/34°C.
fT_photo	0–1	Mean temperature stress factor for photosynthesis (Eq 20, Fig 10). Breakpoints: 0/10/25/35°C.
heat_days	days	Count of days with $T_{\text{max}} > 34^\circ\text{C}$. Heat senescence events (Eq 81, Fig 29).
frost_days	days	Count of days with $T_{\text{min}} < 0^\circ\text{C}$. Frost damage potential (Eq 80).
vpd_mean	kPa	Mean vapour pressure deficit (Eq 88). Drives transpiration efficiency (Eq 87).
fasw_mean	0–2	Mean fraction of available soil water: SM/PAWC (proxy for Eq 70).
fw_photo	0–1	Mean photosynthesis water stress: $\min(\text{FASW}/0.5, 1)$ (proxy for Eq 95–96).
n_days	days	Number of valid days in the slot. Handles partial weeks/months at boundaries.

4 Weekly Features (704 features)

4.1 Slot Naming Convention

Slot name	Meaning
wpre4 ... wpre1	4 weeks before sowing (pre-season)
w0	Week of sowing (days 0–6 from sowing)
w1 ... w39	Weeks 1–39 after sowing

Each slot spans exactly 7 days relative to the per-cell sowing date. Pre-sowing weeks capture antecedent soil moisture recharge and rainfall patterns.

Total: 44 slots $\times$ 16 variables = 704 features.

Example column names: rain_sum_wpre2, fasw_mean_w5, heat_days_w20, tt_sum_w0.

4.2 Approximate Phenological Alignment

Weeks from sowing	Approx. stage	Key processes
w0–w2	Establishment	Germination, emergence
w3–w15	Vegetative	Tillering, leaf expansion, vernalisation
w16–w18	Pre-anthesis	Floral initiation, stem elongation
w19–w28	Grain fill	Grain number set, grain filling
w29–w39	Maturation	Senescence, harvest

Note: the exact alignment varies by location and year because phenological progression depends on accumulated thermal time, not calendar weeks.

5 Monthly Features (192 features)

5.1 Slot Naming Convention

Slot name	Meaning
mpre3 ... mpre1	3 calendar months before the sowing month
m0	Sowing month (only post-sowing days included)
m1 ... m8	Months 1–8 after sowing month

Monthly slots are defined by calendar months relative to each cell’s sowing month. For a May sowing, mpre1 = April, m0 = May (post-sowing days only), m3 = August, etc.

Total: 12 slots $\times$ 16 variables = 192 features.

Pre-sowing months capture soil moisture recharge (e.g. autumn break rainfall in southern Australia), stored soil water, and temperature regime before the crop is established.

Example column names: rain_sum_mpre1, fw_photo_m4, rad_sum_m7.

6 Crop-Seasonal Features (~121 features)

6.1 APSIM Phenological Windows

Features are aggregated over five thermal-time-based windows (same 16 variables as Section 3):

Window	Cum. TT range (°Cd)	Phenological period	Slot suffix
W1_estab	0 – 100	Sowing → Emergence	_W1_estab
W2_veg	100 – 1055	Emergence → Floral Initiation	_W2_veg
W3_preAnth	1055 – 1175	Floral Init. → Flowering	_W3_preAnth
W4_grainFill	1175 – 1860	Flowering → End Grain Fill	_W4_grainFill
W5_matur	1860 – 2060	End Grain Fill → Maturity	_W5_matur
crop	0 – 2060	Whole crop cycle	_crop

Subtotal: 6 windows × 16 variables = 96 features.

Thermal time targets are for the Hartog cultivar (APSIM default). Cumulative thermal time is calculated using Eq 4 with cardinal temperatures 0/26/34°C.

6.2 Additional Crop-Seasonal Features

Feature	Windows	Description (APSIM reference)
<i>Water stress — expansion</i>		
fw_expan_mean_{W}	W2, W3	Mean leaf expansion stress factor. More sensitive than fw_photo: threshold FASW/0.8 (Eq 97, Fig 34).
fw_expan_min_{W}	W2, W3	Minimum (worst) expansion stress in the window.
<i>Water stress — photosynthesis minimum</i>		
fw_photo_min_{W}	W2, W3, W4	Worst single-day photosynthesis water stress. Captures peak drought severity.
<i>Drought event counts</i>		
drought_severe_{W}	W2, W3, W4	Days with fw_photo < 0.5. Severe water stress events.
drought_moderate_{W}	W2, W3, W4	Days with 0.5 ≤ fw_photo < 0.8. Moderate stress events.
<i>Cumulative water stress</i>		
cum_water_stress_crop	Whole crop	∑(1 – fw_photo) over all crop days. Total season water deficit.
cum_water_stress_{W}	W3, W4	Cumulative stress in pre-anthesis and grain fill.
<i>Soil water dynamics</i>		
FASW_at_sowing	Sowing ±5d	Mean FASW at sowing. Initial soil water status.
FASW_W4_drying_rate	W4	FASW _{1st half} – FASW _{2nd half} of grain fill. Soil water depletion rate during grain filling.
<i>Developmental factors</i>		
vern_sum_W2_veg	W2	Cumulative vernalisation (Eq 9–12). Drives phenological progress from emergence to floral initiation.
fD_mean_W2_veg	W2	Mean photoperiod factor (Eq 8). Short days slow development.

Feature	Windows	Description (APSIM reference)
<i>Grain fill factors</i>		
hg_mean_W4	W4	Mean grain fill temperature factor (Eq 49, Fig 17). Optimal at 18–25°C.
heat_sen_W4	W4	Mean heat senescence factor. Non-zero when Tmax > 34°C (Eq 81).
<i>Transpiration and total thermal time</i>		
VPD_mean_crop	Whole crop	Mean VPD (Eq 88). Controls transpiration efficiency.
TE_mean_crop	Whole crop	Mean transpiration efficiency proxy: 0.006/VPD (Eq 87).
TT_cum_crop	Whole crop	Total thermal time accumulated. Indicates season length.

7 APSIM Equation Summary

Key equations from the APSIM-Wheat documentation used in feature computation:

$$\text{Crown temperature: } T_c = \frac{T_{c,\max} + T_{c,\min}}{2} \quad (\text{Eq 3})$$

$$\text{Thermal time: } \Delta TT = \begin{cases} T_c & 0 < T_c \leq 26 \\ \frac{26}{8}(34 - T_c) & 26 < T_c \leq 34 \\ 0 & \text{otherwise} \end{cases} \quad (\text{Eq 4})$$

$$\text{Photoperiod factor: } f_D = 1 - 0.002R_p(20 - L_p)^2 \quad (\text{Eq 8})$$

$$\text{Vernalisation: } \Delta V = \min\left(1.4 - 0.0778T_c, 0.5 + \frac{13.44T_c}{(T_{\max} - T_{\min} + 3)^2}\right) \quad (\text{Eq 9})$$

$$\text{Photosynthesis stress: } f_{T,\text{photo}} = h_{T,\text{photo}}\left(\frac{T_{\max} + T_{\min}}{2}\right) \quad (\text{Eq 20})$$

$$\text{VPD: } VPD = 0.75\left[6.1078 e^{17.269T_{\max}/(237.3+T_{\max})} - 6.1078 e^{17.269T_{\min}/(237.3+T_{\min})}\right] \quad (\text{Eq 88})$$

$$\text{Water stress (photo): } f_{w,\text{photo}} = W_u/W_d \quad (\text{Eq 95})$$

In our proxy implementation, $f_{w,\text{photo}} \approx \min(\text{FASW}/0.5, 1)$ using gridded soil moisture divided by PAWC as a substitute for the full APSIM layered water balance.

8 Notes

- **Yield units:** Wheat yield is converted from kg/ha to t/ha during extraction.
- **Missing values:** Cells with `no_sow = 1` have zeros in all post-sowing features and NaN in yield. Filter these before ML training.
- **NaN in slot features:** `tmax_max` and `tmin_min` are NaN when a slot has zero valid days. Other variables are zero.

- **Temporal overlap:** Weekly and monthly features for the same period are computed independently — they are not nested. XGBoost can select the most informative scale.
- **Pre-sowing features:** These capture antecedent conditions (stored soil water, autumn break rainfall) that influence sowing decisions and early crop establishment.
- **Configuration:** All parameters (paths, year range, thermal time windows, number of weekly/monthly slots) are in `config.py`. Temporal settings can be changed without modifying the extraction code.